SMTP Tester Pro v2.0

Comprehensive User Guide

- Advanced SMTP Credential Testing Tool
- Async Support for Faster Checks
- Multiple Proxy Types (HTTP/SOCKS4/SOCKS5/Rotating)
- Beautiful Terminal UI with Progress Tracking
- Cross-Platform: Windows & Linux Support

Complete Installation & Usage Instructions

Windows | Linux | VirtualBox Setup

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1. Introduction

SMTP Tester Pro is a standalone, advanced SMTP credential testing tool designed for security professionals, system administrators, and penetration testers. This comprehensive tool provides a modern, efficient way to test SMTP server configurations and validate email credentials with support for multiple proxy types and asynchronous operations.

1.1 Key Features

- Standalone Application: No external library dependencies beyond standard Python packages
- Asynchronous Support: Perform multiple tests simultaneously for faster results
- Multi-threaded Execution: Configurable thread pool for parallel testing
- Beautiful Terminal UI: Rich interface with progress bars and colored output
- Multiple Proxy Types: HTTP, SOCKS4, SOCKS5, and rotating proxy support
- SSL/TLS Security: Support for none, SSL, TLS, and STARTTLS connections
- Multiple Export Formats: Export results to TXT, JSON, or CSV formats
- Resume Capability: Skip already tested credentials and resume sessions
- Configuration Files: Support for JSON and YAML configuration files
- Utility Tools: Banner grabbing, MX lookup, and email validation

1.2 System Requirements

Operating Systems:

- Windows 10/11 (64-bit)
- Windows Server 2016/2019/2022
- Ubuntu 20.04/22.04/24.04 LTS
- Debian 10/11/12
- Kali Linux 2023.x
- Any modern Linux distribution with Python 3.8+

Software Requirements:

- Python 3.8 or higher
- pip (Python package manager)
- Internet connection for SMTP testing

1.3 Project Structure

File	Description
smtp_tester.py	Main application file
utils.py	Utility tools (banner grab, MX lookup)
requirements.txt	Python dependencies list
config.json	Sample configuration file
install_windows.bat	Windows auto-installer script
run_interactive.bat	Windows interactive mode launcher
run_quick.bat	Windows quick start wizard
run_utils.bat	Windows utilities menu
install.sh	Linux/macOS installer script
data/emails.txt	Email list input file
data/passwords.txt	Password list input file
data/proxies.txt	Proxy list input file

2. Windows Installation

Windows installation is fully automated with the included batch scripts. The installer will check for Python, download it if necessary, install all dependencies, and create helper scripts.

2.1 Method 1: Automated Installation (Recommended)

1. Download the Project: Extract the SMTP Tester Pro folder to your desired location
2. Run the Installer: Double-click `install_windows.bat`
3. Administrator Prompt: Click 'Yes' if prompted for administrator privileges
4. Python Detection: The script will check if Python is installed
5. Python Installation: If Python is not found, the script will offer to download and install Python 3.12 automatically
6. Virtual Environment: Choose whether to create a virtual environment (recommended)
7. Dependency Installation: All required packages will be installed automatically
8. Sample Files: Sample data files will be created in the `data/` folder
9. Launch: Choose to start SMTP Tester immediately after installation

2.2 Method 2: Manual Installation

1. Install Python: Download from https://www.python.org/downloads/
2. During installation, check 'Add Python to PATH'
3. Open Command Prompt or PowerShell
4. Navigate to the project directory: `cd path\to\smtp_tester`
5. Install dependencies: `pip install -r requirements.txt`
6. Run the application: `python smtp_tester.py`

2.3 Quick Launch Scripts

After installation, use these batch files to launch different modes:

Script	Purpose
run_interactive.bat	Launches SMTP Tester in interactive mode with prompts

run_quick.bat	Guided quick start with server presets and options
run_utils.bat	Utility tools menu (banner grab, MX lookup, etc.)

3. Linux Installation

Linux installation can be done using the provided shell script or manually through the terminal. The following instructions cover various Linux distributions.

3.1 Method 1: Using Install Script

Open a terminal and run the following commands:

```
cd /path/to/smtp_tester
chmod +x install.sh
./install.sh
```

3.2 Method 2: Manual Installation (Ubuntu/Debian)

1. Update package lists: `sudo apt update`
2. Install Python and pip: `sudo apt install python3 python3-pip python3-venv -y`
3. Navigate to project: `cd /path/to/smtp_tester`
4. Create virtual environment (optional): `python3 -m venv venv`
5. Activate virtual environment: `source venv/bin/activate`
6. Install dependencies: `pip install -r requirements.txt`
7. Run the application: `python smtp_tester.py`

3.3 Kali Linux Installation

Kali Linux comes with Python pre-installed. Simply run:

```
cd /path/to/smtp_tester
pip install -r requirements.txt
python smtp_tester.py
```

3.4 Common Linux Issues

- Permission denied: Run `chmod +x install.sh` or use sudo

- pip not found: Install with `sudo apt install python3-pip`
- Externally managed environment: Use virtual environment or `pip install --break-system-packages`
- Module not found: Ensure all dependencies are installed with `pip install -r requirements.txt`

4. VirtualBox Setup for Linux

Running SMTP Tester Pro in a virtual machine provides isolation and security. This section covers the complete setup of Linux on VirtualBox for testing purposes.

4.1 Prerequisites

- Oracle VirtualBox (latest version) - Download from https://www.virtualbox.org/
- Linux ISO image (Ubuntu 22.04 LTS recommended) - Download from https://ubuntu.com/download
- At least 4GB RAM allocated to VM
- At least 25GB virtual hard disk space
- VT-x/AMD-V virtualization enabled in BIOS

4.2 Step 1: Create Virtual Machine

1. Open VirtualBox and click 'New'
2. Enter VM name: 'SMTP-Tester-Linux' or similar
3. Select Type: 'Linux'
4. Select Version: 'Ubuntu (64-bit)' or your chosen distribution
5. Set Memory: At least 4096MB (4GB) recommended
6. Create virtual hard disk: At least 25GB, VDI format, dynamically allocated
7. Click 'Create' to finish VM creation

4.3 Step 2: Configure VM Settings

Select the VM and click 'Settings'. Configure the following:

- System > Processor: Allocate at least 2 CPU cores
- Display: Set Video Memory to 128MB, enable 3D acceleration
- Storage: Attach the Linux ISO to the optical drive
- Network: Set to 'Bridged Adapter' for direct network access, or 'NAT' for basic connectivity
- Shared Folders: Add a shared folder to transfer files between host and guest

4.4 Step 3: Install Linux

1. Start the VM by clicking 'Start'
2. Select the Linux ISO if prompted
3. Follow the installation wizard
4. Choose 'Install Ubuntu' or similar option
5. Select keyboard layout and continue
6. Choose 'Normal installation' with optional third-party drivers
7. Select 'Erase disk and install Ubuntu' (for VM use only)
8. Create user account and set password
9. Wait for installation to complete and restart

4.5 Step 4: Post-Installation Setup

1. Update the system: `sudo apt update && sudo apt upgrade -y`
2. Install VirtualBox Guest Additions for better integration
3. Install Python dependencies: `sudo apt install python3-pip python3-venv -y`
4. Create a working directory: `mkdir ~/smtp_tester`
5. Transfer SMTP Tester Pro files to the VM using shared folders or SCP

4.6 Step 5: Install and Run SMTP Tester Pro

1. Open terminal in the project directory
2. Make install script executable: `chmod +x install.sh`
3. Run installer: `./install.sh`
4. Follow the prompts to complete installation
5. Launch SMTP Tester: `python smtp_tester.py`

4.7 Network Configuration

Mode	Description	Use Case
NAT	VM shares host IP	Basic testing, simple setup
Bridged	VM gets own IP on network	Full network access, recommended

Host-Only	VM only accessible from host	Isolated testing environment

5. Quick Start Guide

This section provides quick instructions to get SMTP Tester Pro running immediately after installation.

5.1 Prepare Input Files

Email List (data/emails.txt):

Add one email address per line. Lines starting with # are comments.

Password List (data/passwords.txt):

Add one password per line. Common passwords work well for testing.

Proxy List (data/proxies.txt) - Optional:

Add proxies in format: host:port or protocol://host:port

5.2 Interactive Mode (Easiest)

Simply run without arguments and follow the prompts:

```
python smtp_tester.py
```

The interactive mode will guide you through:

- SMTP server configuration (host, port, security)
- Input file selection
- Proxy configuration (optional)
- Performance settings
- Confirmation before starting

5.3 Quick Command Line

For Gmail testing:

```
python smtp_tester.py -H smtp.gmail.com -P 587 -e data/emails.txt -p
data/passwords.txt
```

For Outlook testing:

```
python smtp_tester.py -H smtp-mail.outlook.com -P 587 -e
data/emails.txt -p data/passwords.txt
```

With proxy:

```
python smtp_tester.py -H smtp.gmail.com -P 587 -e data/emails.txt -p
data/passwords.txt --proxy-type socks5 --proxy-list data/proxies.txt
```

5.4 Common SMTP Server Settings

Provider	Host	Port	Security
Gmail	smtp.gmail.com	587	STARTTLS
Gmail (SSL)	smtp.gmail.com	465	SSL
Outlook	smtp-mail.outlook.com	587	STARTTLS
Yahoo	smtp.mail.yahoo.com	587	STARTTLS
iCloud	smtp.mail.me.com	587	STARTTLS
Office 365	smtp.office365.com	587	STARTTLS

6. Command Line Reference

SMTP Tester Pro supports extensive command line options for advanced usage and automation.

6.1 SMTP Settings

Argument	Description	Default
-H, --host	SMTP server hostname	Required
-P, --port	SMTP server port	587
--security	Security mode: none, ssl, tls, starttls	starttls
--timeout	Connection timeout in seconds	30

6.2 Input Files

Argument	Description	Default
-e, --emails	Path to email list file	emails.txt
-p, --passwords	Path to password list file	passwords.txt

6.3 Proxy Settings

Argument	Description	Default
--proxy	Single proxy (protocol://host:port)	None
--proxy-list	Path to proxy list file	None
--proxy-type	Type: none, http, socks4, socks5, rotating	none

6.4 Performance Settings

Argument	Description	Default
-w, --workers	Number of worker threads	10

--async-tasks	Number of async tasks	50
--rate-limit	Requests per second (0=unlimited)	0
--retry	Number of retry attempts	3
--no-async	Disable async mode	False

6.5 Output Settings

Argument	Description	Default
-o, --output	Output directory	results
-f, --format	Output format: txt, json, csv, all	txt
--no-save-tried	Don't save tried credentials	False
--no-resume	Don't resume from previous session	False

7. Configuration Files

SMTP Tester Pro supports configuration files in JSON and YAML formats. This allows you to save settings and reuse them across multiple sessions.

7.1 Using Configuration Files

Load configuration from file:

```
python smtp_tester.py --config config.json
```

Create default configuration file:

```
python smtp_tester.py --create-config my_config.json
```

7.2 JSON Configuration Example

```
{ "email_file": "data/emails.txt", "password_file": "data/passwords.txt", "output_dir":
"results", "smtp_host": "smtp.gmail.com", "smtp_port": 587, "smtp_security":
"starttls", "smtp_timeout": 30, "proxy_type": "none", "max_workers": 10,
"max_async_tasks": 50, "rate_limit": 0, "retry_count": 3, "retry_delay": 1.0,
"output_format": "txt", "verbose": false, "log_level": "info", "save_tried": true,
"resume": true }
```

7.3 Configuration Options Reference

Option	Type	Description
email_file	string	Path to email list file
password_file	string	Path to password list file
output_dir	string	Output directory for results
smtp_host	string	SMTP server hostname
smtp_port	integer	SMTP server port
smtp_security	string	Security mode (none/ssl/tls/starttls)
smtp_timeout	integer	Connection timeout in seconds

proxy_type	string	Proxy type (none/http/socks4/socks5/rotating)
max_workers	integer	Number of worker threads
max_async_tasks	integer	Number of async tasks
rate_limit	integer	Rate limit (0=unlimited)
retry_count	integer	Number of retry attempts
output_format	string	Output format (txt/json/csv/all)
verbose	boolean	Enable verbose output
save_tried	boolean	Save tried credentials
resume	boolean	Resume from previous session

8. Proxy Configuration

SMTP Tester Pro supports multiple proxy types for distributing requests and maintaining anonymity. Proxy usage is completely optional but recommended for large-scale testing.

8.1 Supported Proxy Types

Type	Description	Best For
HTTP	Standard HTTP proxy	Basic anonymity, web proxies
SOCKS4	SOCKS4 protocol	TCP connections, no auth
SOCKS5	SOCKS5 protocol with auth support	Most compatible, recommended
Rotating	Automatic rotation through proxy list	Large-scale testing, avoid blocks

8.2 Proxy List File Format

Create a file with one proxy per line:

- Basic format: host:port
- With protocol: protocol://host:port
- With authentication: protocol://host:port:username:password
- Comments: Lines starting with # are ignored

```
# SOCKS5 Proxies 127.0.0.1:1080 192.168.1.100:9050 # HTTP Proxies
http://proxy.example.com:8080 # With Authentication
socks5://proxy.example.com:1080:user:pass
```

8.3 Using Proxies

Single proxy from command line:

```
python smtp_tester.py --proxy socks5://127.0.0.1:1080 ...
```

Proxy list from file:

```
python smtp_tester.py --proxy-type socks5 --proxy-list data/proxies.txt
...
```

8.4 Proxy Statistics

SMTP Tester Pro tracks proxy performance during testing. After completion, you can view statistics showing the number of attempts, successes, failures, and success rate for each proxy. This helps identify the best-performing proxies for future use.

9. Utility Tools

SMTP Tester Pro includes several utility tools for SMTP analysis and email validation. These tools are available in the utils.py module and through the run_utils.bat script.

9.1 SMTP Banner Grabber

Retrieves SMTP server banner and identifies server type:

```
python utils.py banner smtp.gmail.com -p 587
```

- Server type detection (Postfix, Exim, Exchange, etc.)
- STARTTLS support detection
- Authentication methods listing
- Response time measurement

9.2 MX Record Lookup

Finds mail servers for a domain:

```
python utils.py mx gmail.com
```

This tool queries DNS MX records and resolves IP addresses for each mail server.

9.3 Email Validator

Validates email addresses and optionally verifies mailbox existence:

```
python utils.py validate user@example.com
python utils.py validate user@example.com --verify
```

- Email format validation
- MX record existence
- Disposable email detection
- Mailbox existence verification (optional)

9.4 SMTP Port Scanner

Scans common SMTP ports to find services:

```
python utils.py scan mail.example.com
python utils.py scan localhost -p 25 587 465 2525
```

Scans for SMTP services on common ports (25, 587, 465, 2525 by default).

10. Troubleshooting

This section covers common issues and their solutions when using SMTP Tester Pro.

10.1 Connection Issues

- Connection Timeout: Increase timeout with --timeout 60
- Connection Refused: Verify the SMTP server address and port are correct
- Network Unreachable: Check your network connection and firewall settings
- SSL Errors: Try different security mode (--security ssl or --security starttls)

10.2 Authentication Issues

- All attempts fail: Email addresses may have 2FA enabled - use app-specific passwords
- Account locked: Too many failed attempts - wait before retrying
- Invalid credentials: Verify email format and check for typos in password list
- App Password required: Gmail and others require app-specific passwords when 2FA is enabled

10.3 Proxy Issues

- Proxy connection failed: Verify proxy is running and accessible
- Authentication failed: Check proxy username and password format
- All proxies fail: Try a different proxy type or verify proxy list format
- Slow performance: Reduce number of workers or use faster proxies

10.4 Python/Installation Issues

- Module not found: Run pip install -r requirements.txt
- Externally managed environment: Use virtual environment or --break-system-packages flag
- Permission denied: Use sudo on Linux or run as administrator on Windows
- Wrong Python version: Ensure Python 3.8 or higher is installed

10.5 Windows-Specific Issues

- Python not found: Ensure Python is added to PATH during installation
- Script won't run: Run install_windows.bat as administrator
- Antivirus blocking: Add exception for the SMTP Tester folder
- Execution policy: Run 'Set-ExecutionPolicy RemoteSigned' in PowerShell

10.6 Linux-Specific Issues

- Permission denied: Run chmod +x on scripts or use sudo
- pip not found: Install with 'sudo apt install python3-pip'
- Rich not displaying correctly: Ensure terminal supports ANSI colors
- Virtual environment issues: Install python3-venv package

10.7 Security Considerations

- Only test accounts you own or have explicit permission to test
- Use rate limiting to avoid triggering security measures
- Consider using proxies to distribute requests
- Ensure compliance with local laws and regulations
- Do not use for illegal activities